

AHTOPOL, Bulgaria

WMO: 156610

Lat: 42.100N

Lon: 27.940E

Elev: 19

StdP: 101.10

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	-3.8	-9.2	1.7	-3.5	-7.3	2.0	-2.0	14.3	-0.5	12.5	1.2	5.4	320	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	29.8	22.2	28.5	22.2	27.4	21.8	24.2	27.7	23.3	27.1	22.6	26.5	3.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.0	26.6	22.0	16.7	26.0	21.1	15.8	25.5	73.3	27.7	69.6	27.3	66.6	26.7	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	9.5	8.1	-8.3	32.9	2.8	2.3	-10.3	34.6	-12.0	35.9	-13.6	37.2	-15.6	38.9		
			-9.1	25.3	2.8	1.3	-11.1	26.2	-12.7	27.0	-14.2	27.7	-16.3	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	4.3	4.7	7.8	11.0	15.8	20.3	22.9	23.3	19.5	15.0	10.3	6.0
	DBStd	7.62	4.60	4.79	4.51	3.52	3.19	2.60	1.86	2.04	2.36	3.46	3.84	4.70
	HDD10.0	658	186	159	99	27	1	0	0	0	0	4	41	142
	HDD18.3	2217	436	382	326	221	90	11	0	0	15	113	241	382
	CDD10.0	1924	8	11	33	57	182	308	401	412	286	158	51	18
	CDD18.3	441	0	0	1	2	13	68	142	153	52	9	1	0
	CDH23.3	2698	0	0	3	9	59	373	956	1096	163	39	1	0
	CDH26.7	444	0	0	0	1	9	58	158	200	10	9	0	0
Wind	WSAvg	3.4	3.6	3.7	3.6	3.0	2.8	2.7	3.1	3.2	3.6	3.8	3.3	3.8
Precipitation	PrecAvg	631	61	52	54	43	34	35	31	53	75	76	84	
	PrecMax	1108	169	126	153	112	84	86	134	131	169	217	187	169
	PrecMin	388	4	9	11	3	4	1	0	1	1	8	4	19
	PrecStd	171	41	31	34	27	23	25	30	39	51	55	49	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	18.9	22.6	24.4	27.6	31.0	31.3	32.5	28.2	27.2	22.9	18.8
		MCWB	12.0	12.1	13.5	14.7	19.5	21.4	22.6	22.3	20.1	18.2	15.2	13.4
	2%	DB	15.9	16.2	19.5	21.3	24.7	27.9	29.6	30.0	26.1	23.5	20.1	17.1
		MCWB	11.2	11.3	12.5	13.5	18.0	21.3	22.3	22.8	20.1	18.2	15.4	12.5
	5%	DB	13.1	14.0	17.0	18.6	22.7	26.5	28.3	28.6	25.1	21.5	17.7	14.9
		MCWB	9.3	10.1	11.3	12.6	17.4	20.7	22.5	22.6	19.9	17.3	14.2	11.4
	10%	DB	10.9	11.7	14.8	16.4	21.1	25.4	27.2	27.5	24.1	19.9	15.9	12.7
		MCWB	8.0	8.7	10.2	12.0	16.7	20.3	21.9	22.2	19.2	16.5	13.4	10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	13.1	14.3	15.7	20.5	23.4	25.2	25.9	22.6	20.0	16.9	14.1
		MCDB	16.7	17.7	21.5	21.6	25.4	28.2	27.9	28.6	25.8	23.9	20.3	17.6
	2%	WB	11.2	11.4	12.9	14.1	19.2	22.3	24.2	24.5	21.6	18.6	15.7	13.0
		MCDB	15.4	15.6	18.3	19.7	23.4	26.4	27.3	27.9	24.6	22.4	19.0	16.3
	5%	WB	9.7	10.2	11.7	13.1	18.1	21.5	23.3	23.6	20.8	17.7	14.7	11.8
		MCDB	12.8	13.7	16.3	17.8	21.7	25.6	27.1	27.2	23.9	21.0	17.4	14.5
	10%	WB	8.3	8.8	10.5	12.2	17.1	20.8	22.5	22.8	19.9	17.0	13.4	10.4
		MCDB	10.4	11.7	14.5	15.9	20.4	24.7	26.6	26.6	23.1	19.7	15.9	12.7
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.8	6.3	6.5	7.3	7.8	7.6	7.7	7.0	6.1	6.2	5.6
		MCDBR	8.6	9.7	9.8	10.4	10.0	9.5	8.8	9.0	8.6	9.1	8.6	8.5
		MCWBR	5.8	5.9	5.1	5.0	5.6	5.3	5.4	5.2	5.0	5.1	5.3	5.6
	5% WB	MCDBR	8.3	9.0	9.1	9.0	8.5	8.6	7.8	7.9	7.4	7.3	7.2	7.5
		MCWBR	5.9	5.7	5.2	4.9	5.4	5.5	5.3	5.2	5.0	4.8	4.8	5.3
Clear-Sky Solar Irradiance	taub	0.365	0.397	0.432	0.477	0.444	0.443	0.448	0.457	0.440	0.413	0.386	0.371	
	taud	2.316	2.216	2.145	2.037	2.213	2.260	2.260	2.230	2.244	2.284	2.297	2.293	
	Ebn at Noon	756	787	804	796	836	835	826	806	789	762	722	708	
	Edh at Noon	88	113	136	163	140	134	133	133	123	105	89	84	
All-Sky Solar Radiation	RadAvg	1.44	2.15	3.42	4.84	6.15	6.78	6.92	6.12	4.40	2.80	1.73	1.27	
	RadStd	0.21	0.26	0.32	0.49	0.39	0.46	0.34	0.29	0.37	0.28	0.23	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	+0.54	N/A	N/A	N/A	+0.51	+0.58	-70	-112	+109	+60

Nomenclature: See separate page